Optimize hospital capacity planning with length-of-stay prediction at admission

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1 Introduction

Hospital bed congestion and emergency department (ED) crowding create severe operational bottlenecks, leading to extended boarding times, ambulance diversions, and cancelled elective procedures (Olshaker and Rathlev [2006]). While capacity management depends on regulating patient admissions and predicting downstream bed availability, operational models often struggle due to limited visibility into how long patients will remain in the hospital.

Prior operations research addresses this bottleneck primarily through admission-side gating and queue control. Helm et al. [2011] formulated Markov Decision Process (MDP) models to establish admission thresholds that balance emergency and scheduled patient flows. Yuan and Su [2026] extend this approach to empirical settings, employing dynamic threshold downgrading policies to automate bed allocation and minimize high-risk patient rejections during demand surges. At the point of intake, decision models using logistic regression have also been proposed to predict admission likelihood at triage, enabling early bed requests to reduce boarding delays.

However, these frameworks typically treat inpatient stay duration as static or aggregate averages. In practice, length of stay (LoS) is highly variable and clinically heterogeneous. To bridge this gap, this project develops supervised machine learning models to predict inpatient Length of Stay at the time of admission. By estimating individual discharge horizons using baseline clinical and demographic intake data, our goal is to replace aggregate assumptions with granular, patient-level duration forecasts and provide hospital administrators with the actionable visibility needed for proactive bed turnover, dynamic capacity planning, and resource allocation.

2 Data

The dataset we are using is the Statewide Planning and Research Cooperative System (SPARCS) Inpatient De-identified File provided by the New York State Department of Health [2024] collected during 2024. It contains information on the recorded facilities, inpatients' demographics, admission types, clinical conditions, services performed, and payments details. Publicly available, the data does not contain any protected health information (PHI) under the Health Insurance Portability and Accountability Act (HIPAA), nor does it contain any individually identifiable data element (Socrata).

2.1 Overview

There are 2,196,737 rows and 33 columns in the raw data. Each row represents a hospital inpatient discharge record and each column describes a piece of inpatient information. The column `los` is our target variable, and is derived from the column `length_of_stay`, but with the value `120+`, representing more than 120 days of stay, converted as `120` (120 days of stay). In this project and for training purposes, we assume that the `ccsr_procedure_code` is known at the time of admission, which is a limitation that we acknowledge may lead to more optimistic predictions for emergency admissions. The final dataset we would be training on contains 1,993,483 rows and 15 columns, including 1 target variable and 14 features.

2.2 Features for Training

The 14 features we would be using for model training are:

- `permanent_facility_id`
- `zip_code`
- `age_group`
- `gender`
- `race`
- `ethnicity`
- `type_of_admission`
- `emergency_department_indicator`
- `ccsr_diagnosis_code`
- `ccsr_procedure_code`
- `apr_medical_surgical`
- `payment_typology_1`
- `payment_typology_2`
- `payment_typology_3`

from which we grouped into the following:

Group	**Features**
Facility / Geography	`permanent_facility_id`, `zip_code`
Patient	`age_group`, `gender`, `race`, `ethnicity`
Admission	`type_of_admission`, `emergency_department_indicator`
Clinical	`ccsr_diagnosis_code`, `ccsr_procedure_code`, `apr_medical_surgical`
Payer	`payment_typology_1`, `payment_typology_2`, `payment_typology_3`

Table 1: Predictors By Groups.

2.3 Dropped Columns

The rest of the columns are dropped for reasons such as data leakage, lack of relevance, and redundancy. More details and explanations are provided in Table 2 (on page 3).

2.4 Dropped Rows

Some rows are also dropped due to usability and relevancy issues. More details and explanations are provided in Table 3 on page 3.

Columns Dropped	Reason	Justification
`discharge_year`	Constant	Only one value (2024), so it cannot explain length of stay.
`patient_disposition`, `total_charges`, `total_costs`	Leakage by definition	They describe the completed stay and are not known at admission. Charges and costs grow with every extra day.
`apr_severity_of_illness_code`, `apr_risk_of_mortality`, `apr_drg_code`	Condition at discharge	The 3M APR-DRG Methodology Overview ([3M Health Information Systems, 2023, p. 53]) says they represent the condition at discharge, including conditions that developed during the stay.
`facility_name`	Duplicate of another column (multicollinearity)	Can be determined by `permanent_facility_id`.
`apr_severity_of_illness`, `apr_mdc_description`, `apr_drg_description`, `ccsr_diagnosis_description`, `ccsr_procedure_description`	1:1 duplicate of a code column (multicollinearity)	Same information as the code column, written as text, so each pair is perfectly collinear.
`length_of_stay`	Replaced by target	Replaced by the numeric `los`.
`birth_weight`	Out of scope	Filled only for newborns (99.97% of newborns). Newborns are a separate cohort, so it is blank for everyone else.
`hospital_county`, `health_service_area`, `operating_certificate_number`	Nested under facility (multicollinearity, Cramér's V ≈ 1.0)	`permanent_facility_id` determines all three, so they add no information and would be collinear with it.
`apr_mdc_code`	Coarser version of diagnosis (multicollinearity, $V = 0.9$)	MDC is defined from the principal diagnosis ([3M Health Information Systems, 2023, p. 57]), so `ccsr_diagnosis_code` already carries it and the two are strongly collinear.

Table 2: Columns Dropped and Justifications

Rows	Count	Reason
Redacted facility records	5,295	Deliberate de-identification (mostly abortion-related). Imputing a facility would create a false signal.
Newborn admissions	197,959	A birth in the hospital is a different process from the one being predicted. Set aside as `df_newborn`.
Ungroupable records	381	Severity 0, mortality Undetermined and medical/surgical Not Applicable. These values are not on the ordered scales and cannot be imputed.

Table 3: Rows Dropped and Justifications

2.5 Missing Values

Missing values in these columns carry meaningful information, so we assign labels based on the underlying reasons for the missingness, as shown in Table 4.

Column	Label	Meaning
`zip_code`	Suppressed	Blanked by SPARCS for small populations, small cells or abortion. Kept separate from `OOS`.
`payment_typology_1`	Not recorded	First payer left blank.
`payment_typology_2`	No second payer	No additional payer.
`payment_typology_3`	No third payer	No additional payer.
`ccsr_procedure_code`	No procedure recorded	Almost all on medical stays, none on surgical stays.

Table 4: Labels for Missing Values

3 Exploratory Data Analysis

3.1 Target Variable

We begin our analysis from the target variable. 1.9% of discharges are stays longer than 30 days, so about 98% last 30 days or fewer. Upon that, most of the stays are short. About half of all discharges end within 3 days: 17.7% at 1 day, 18.1% at 2 days, 15.1% at 3 days. The distribution is strongly right-skewed. The mean of 6.04 days is about twice the median of 3 days, and the 95th percentile is 19 days, as shown in Figure 1.

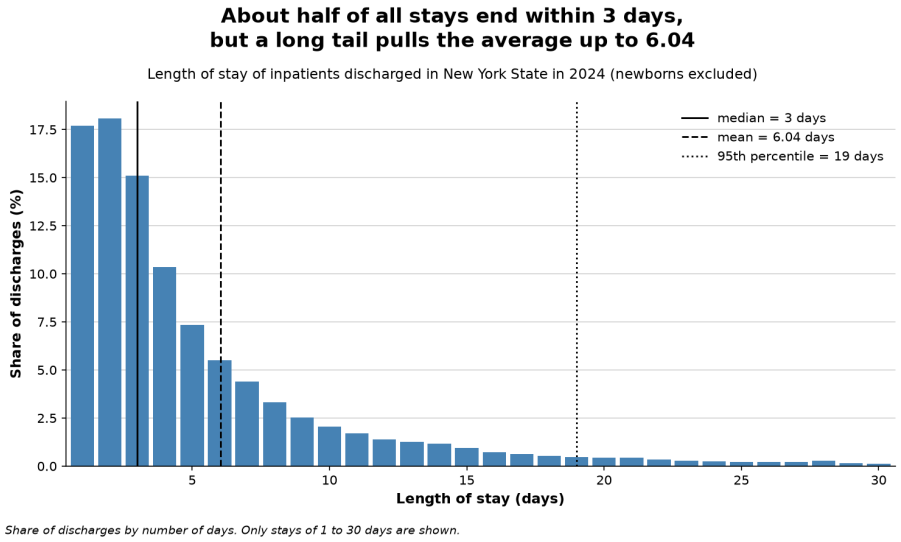


Figure 1: Share of discharges by number of days (Only stays of 1 to 30 days are shown)

3.2 Categorical Analysis

We next examine the feature variables to better understand the characteristics of the patient population and admission patterns. The dataset is skewed toward older patients, with 35.5% of admissions involving patients aged 70 or older and another 29.1% involving patients aged from 50 to 69. In contrast, minors aged from 0 to 17 account for only 5.2% of all admissions. Emergency

admissions predominate, accounting for 73.1% of admissions, and 68.9% of patients pass through the emergency department (ED). Medicare (46.4%) and Medicaid (26.8%) dominate the primary payer. Just over half (52.5%) have no second payer, and 88.6% of the discharges have no third payer.

3.3 Bivariate Analysis

We then move on to analyze the relationship between each feature and the target variable. In Figure 2, we could see that the diagnosis separates the typical stay strongly: among the 15 most common diagnoses the median runs from about 2 days to about 13 days. It also changes the spread of the stays: the middle half of schizophrenia stays runs from about 7 to 21 days, while for osteoarthritis and cardiac dysrhythmias the stays are only about 1 to 4 days.

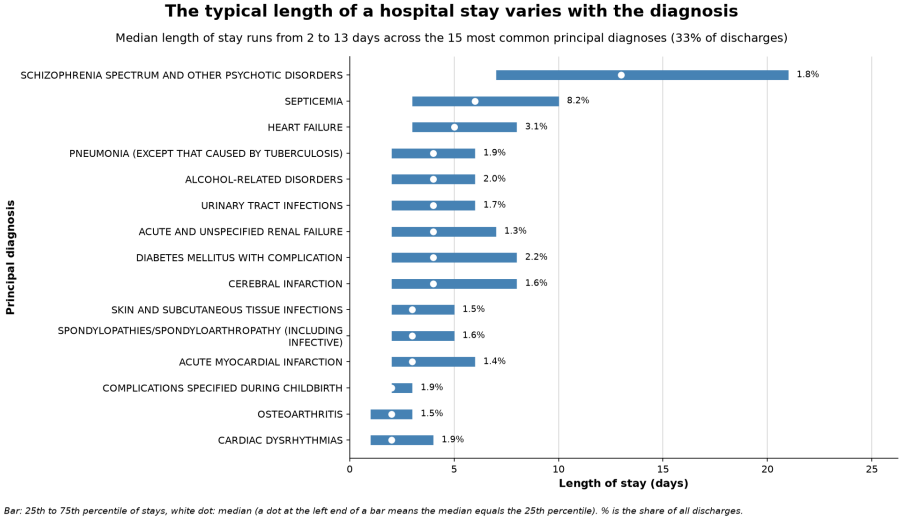


Figure 2: Principle Diagnosis by number of days

Moreover, among the 12 most common procedure categories, typical lengths of stay range from approximately 2 days for spontaneous vaginal delivery to 7 days for venous and arterial catheter placement, with the middle 50% of stays for the latter ranging from 4 to 13 days. Delivery-related procedures, including spontaneous vaginal delivery and cesarean section, have the shortest stays (2-3 days), whereas catheter placement, non-invasive ventilation, and blood transfusion are associated with the longest stays (5-7 days).

We also examined the potential impact of selected features with smaller cardinalities. As shown in Figure 3, the median stay increases with age, from 2 days among patients aged from 0 to 17 to 4 days among those aged from 50 to 69 and 70 or older. By primary payer, the median stay is 3 days for most payer groups, compared with 4 days for patients covered by Medicare or the Department of Corrections. Similarly, the median stay is 3 days for most racial groups and 4 days for Black/African American patients. Patients with an ED visit also have a longer median stay of 4 days, compared with 3 days for those without an ED visit. However, the distributions of length of stay overlap substantially across all groups, suggesting that no single demographic or admission-related characteristic is sufficient on its own to distinguish long stays from the short ones.

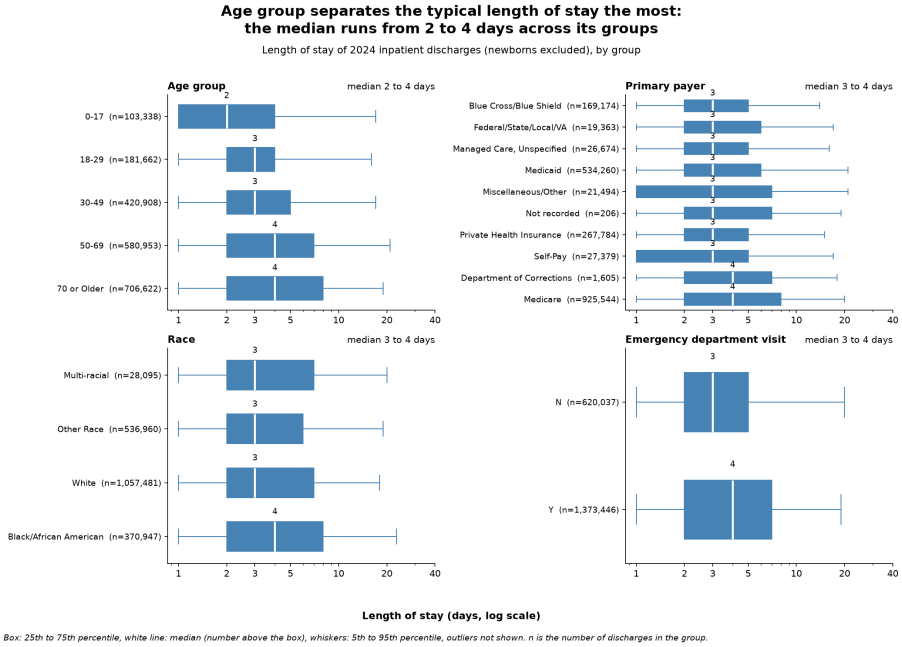


Figure 3: Principle Diagnosis by number of days

4 Problem description and potential impact

[TBD after following up question] I am quite confused on the expectation of this section because I think I have done similar work in the introduction. To recap, in the introduction, I introduced the problem (problem description), where are the gaps that our work might be helpful, and how our work will help improve the situation (potential impact). Should we be more direct and details in this section? How should we avoid repeating our work the introduction or it is expected to repeat ourselves a bit here?

The immediate impact of the project is being able to predict the length of the time a patient would be occupying an inpatient bed given the information known at the admission (or the updated information during the patient's time of stay before being discharged). Derived potential impacts include:

- more efficient and timely bedding, staffing and resources planning for the coordinators and hospital operation teams;
- earlier discharge planning, such as arrangements of the rehabilitation facilities, home care or post-acute care, for the families and care providers;
- more informed resource allocation and planning in the long-term for hospital administrators;
- more informed expectations for the patients themselves.

References

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